Sports Medicine
https://doi.org/10.1007/s40279-020-01302-y

ORIGINAL RESEARCH ARTICLE

Medical Disqualification Following Concussion in Collegiate Student-Athletes: Findings from the CARE Consortium

Julianne D. Schmidt[1,2] · Michelle L. Weber Rawlins[3] · Robert C. Lynall[1,2] · Christopher D'Lauro[4] · James R. Clugston[5] · Tom W. McAllister[6] · Michael McCrea[7] · Steven P. Broglio[8,9] · CARE Consortium Investigators

© Springer Nature Switzerland AG 2020

Abstract

Background The absence of evidence-based guidelines make medical disqualification (MDQ) following concussion one of the most challenging decision-making processes faced by sports medicine professionals.

Objective We aimed to compare premorbid and postmorbid factors between student-athletes that were and were not medically disqualified from sport following a concussion.

Methods Among 1832 student-athletes diagnosed with concussion within the CARE Consortium, 53 (2.9%) were medically disqualified (MDQ+) and 1779 (97.1%) were not medically disqualified (MDQ−). We used contingency tables and descriptive statistics for an initial evaluation of a broad list of premorbid and postmorbid factors. For those factors showing association with MDQ status, we calculated odds ratios and 95% confidence intervals for the odds of being MDQ+in the presence of the identified factor.

Results History of 2 (OR: 3.2, 95% CI 1.5, 6.9) or 3+(OR: 7.4, 95% CI 3.4, 15.9) previous concussions; 1+headaches in past 3 months (OR: 1.8, 95% CI 1.0, 3.2); immediate removal from play (OR: 2.4, 95% CI 1.2, 4.9); alcohol (OR: 2.6, 95% CI 1.2, 5.4), tobacco (OR: 3.3, 95% CI 1.1, 9.5), or marijuana use since injury (OR: 5.4, 95% CI 1.5, 19.0); as well as prolonged recovery due to mental health alterations (OR: 5.3, 95% CI 2.0, 14.1) or motivation/malingering (OR: 7.5, 95% CI 3.3, 17.0) increased odds of being MDQ+. The MDQ+group took longer to become asymptomatic relative to the MDQ− group (MDQ+: 23.5 days, 95% CI 15.8, 31.2; MDQ−: 10.6 days, 95% CI 9.5, 11.6; $p<0.001$).

Conclusions MDQ following concussion was relatively rare. We identified three patterns related to MDQ following concussion: (1) concussion and headache history were the only premorbid factors that differed (2) initial concussion presentation was more severe and more immediate in the MDQ+group, and (3) post-concussion recovery outcomes expressed the greatest differences between groups.

Key Points

Medical disqualification after concussion was considerably rare in this large-scale study of NCAA student-athletes.

Overall, medical disqualification following concussion was more prevalent in athletes with more extensive concussion histories, more frequent premorbid headaches, more severe initial presentations, that then experienced a longer and more complicated recovery.

Our findings further highlight that each concussion case presents and is treated individually and that no strict parameters can currently be applied by all clinicians when making difficult MDQ decisions.

This article is part of a Topical Collection on The NCAA-DoDConcussion Assessment, Research and Education (CARE) Consortium.

The members of the CARE Consortium are listed in acknowledgements section.

Electronic supplementary material The online version of this article (https://doi.org/10.1007/s40279-020-01302-y) contains supplementary material, which is available to authorized users.

✉ Julianne D. Schmidt
schmidtj@uga.edu

Extended author information available on the last page of the article

Published online: 16 June 2020

1 Introduction

Medical disqualification (MDQ) following concussion is one of the most challenging decision-making processes faced by sports medicine professionals [1]. Concussions are among the most common MDQ injuries in a wide range of high school sports including boys' football, soccer, ice hockey, and lacrosse as well as girls' softball, basketball, field hockey, lacrosse, and soccer [2]. MDQ injuries among collegiate athletes have not been studied.

Typically, MDQ decisions are made with concern over long-term health and consideration of severe potential outcomes of continued sport participation. In regards to MDQ following concussion, the American Medical Society for Sports Medicine (AMSSM) position statement states "There are no evidence-based guidelines for disqualifying or retiring an athlete from sport after concussion; therefore, each athlete should be carefully and individually assessed to determine the safety and potential long-term health consequences of continued participation" [3]. The following factors have been proposed as considerations for MDQ following concussion: structural abnormality on neuroimaging, multiple lifetime concussions, persistent diminished academic or workplace performance, persistent postconcussive symptoms, prolonged recovery courses, and perceived reduced threshold of sustaining recurrent concussions [4, 5]. However, these recommendations have been based predominantly on clinical opinion [4, 5] without any quantification of relevant factors. This is particularly challenging given the uncertainty of predicting long-term neurological outcomes from contemporary clinical recovery and symptoms.

Sports medicine professionals faced with MDQ decisions often consider multiple individual factors from the student-athlete's medical and academic history, but also consider post-injury characteristics such as duration of recovery and level of impairment [1]. Beyond these medical factors, health care providers also face a wide variety of psycho-social considerations. Collegiate student-athletes have often devoted much of their life to their sport training and often strongly identify with their sport participation [6]. Future professional plans and academic aptitude are also commonly cited clinical considerations. Some athletes aspire to play professionally, while others may be more focused on future professional achievements outside of sport. Because of this level of complexity, the process of MDQ decision-making is difficult. Medically disqualifying a student-athlete carries a potentially heavy burden, often leaving the clinician managing a challenging psycho-social issue in addition to a medical one. Describing the current demographic (premorbid) and injury characteristics (postmorbid) among collegiate student-athletes that are medically disqualified following concussion may help practicing clinicians better understand current clinical practices surrounding this complex decision-making process. We aimed to compare premorbid and postmorbid factors between student-athletes that were and were not medically disqualified (MDQ) from participation in sport following a concussion. We hypothesized that grade point average, sport contact level, NCAA division, starting position status, concussion history, and prolonged recovery would be associated with MDQ status.

2 Methods

2.1 Participants

Data were collected as part of the Concussion Assessment, Research, and Education (CARE) Consortium study that includes varsity student-athletes and military cadets from 26 universities and three military service academies (29 sites total). We excluded non-varsity athlete military cadets, because the MDQ process differs from that involving NCAA student-athletes. Methods for the CARE Consortium have previously been described in further detail [7]. Each participant completed a baseline demographics questionnaire, symptom checklist, a mental status exam, a computerized neurocognitive assessment, and a balance assessment. If injured, participants completed the same assessments again at serial time points following injury. Managing clinicians completed case report forms detailing the injury and recovery immediately following the injury and again at return to play or at the time of MDQ decision. All participants signed informed consent forms approved by each site's institutional review board and the United States Army Human Research Protection Office. This study was conducted in accordance with the standards of ethics outlined in the Declaration of Helsinki.

2.2 Determining MDQ Status

Medical disqualification status was determined using the responses to the following question: "When did the athlete return to unrestricted play?". Concussion cases were categorized as MDQ + if the clinician selected "athlete disqualified for the season", "athlete disqualified/retired from sport for post-concussion syndrome", or "athlete disqualified/retired from multiple concussions". It is important to note that none of these response options indicate whether MDQ status was permanent. Cases were considered MDQ− if the clinician selected "the day symptoms resolved", "following a return to play progression", or "while symptomatic". Since MDQ decisions are highly individualized, we chose to use most recent concussion in cases where participants had more than one injury in the dataset. This is also necessary to fulfill assumptions of independence for our analyses.

2.3 Premorbid and Postmorbid Factors

A full list of factors compared between MDQ + and MDQ− are listed in Tables 1, 2, 3, 4, 5, 6. We compared demographic and scholastic (Table 1), sport (Table 2), medical history (Table 3), injury event (Table 4), concussion presentation (Table 5), and post-concussion outcomes (Table 6) between groups. Some factors represent survey data, such as the Brief Sensation Seeking Scale [8], Sport Concussion Assessment Tool 3 Symptom checklist [9], and the Brief Symptom Inventory 18 [9].

2.4 Statistical Analysis

Statistical analyses were conducted with SPSS (v25.0; SPSS Inc, Chicago, IL). Due to the exploratory nature of this study, we used an a priori alpha of < 0.05 and < 0.10 to discuss trends. We first ran descriptive analyses across all outcome variables. For categorical outcomes, we compared distributions across the MDQ + and MDQ− groups using chi-square tests of association or Fisher's exact tests based on cell counts. Continuous variables were first assessed for distribution normality using Shapiro–Wilk tests. Accordingly, we used either a one-way ANOVA or non-parametric Kruskal–Wallis, depending on data normality, to compare MDQ + and MDQ− groups. We took a liberal approach with our initial exploration of premorbid and postmorbid factors due to the rare nature of MDQ following concussion. We supplement significant or trending descriptive analyses with odds ratios and 95% confidence intervals. Odds ratios were considered significant if the 95% confidence interval did not contain 1.0 (Fig. 1). We present a survival curve for overall MDQ across days to asymptomatic (Fig. 2). Kaplan–Meier Survival curve analyses across factors associated with MDQ are presented as electronic supplemental material.

3 Results

We had a final sample of 1832 unique student-athlete injuries. We excluded 198 concussion cases that preceded the most recent concussion case of interest. Of the 1832 concussed NCAA student-athletes, 53 (2.9%) were MDQ + and 1779 (97.1%) were MDQ−. Among the 53 MDQ + cases, 41.5% ($n = 22$) were disqualified for the season, 22.6% ($n = 12$) were disqualified from sport for post-concussion syndrome, and 35.9% ($n = 19$) were disqualified from sport for multiple concussions.

3.1 Premorbid Factors

Tables 1, 2 and 3 detail the associations between MDQ status and demographic and scholastic (Table 1), sport (Table 2), and medical history (Table 3). No demographic, scholastic, sport, or injury event factors were significantly associated with MDQ status. Among medical history factors, student-athletes with 2 and 3 + concussions prior to the studied concussive event had 3.2 (95% CI 1.5, 6.86) and 7.4 (95% CI 3.4, 15.9) greater odds of being MDQ + , respectively (Fig. 1). Student-athletes who endorsed experiencing one or more headaches in past 3 months (at baseline) had 1.8 (95% CI 1.0, 3.2) greater odds of being MDQ + (Fig. 1). The MDQ + groups trended towards higher sensation seeking on the Brief Sensation Seeking Scale ($p = 0.050$) (Table 3). We also observed trends such that the MDQ + group had slightly lower prevalence of expecting to be a starter ($p = 0.093$), lower prevalence of expectations to be 1st or 2nd string on the depth chart ($p = 0.077$), and fewer years as a starter in their primary position ($p = 0.082$).

3.2 Postmorbid Factors

Tables 4, 5 and 6 detail the associations between MDQ status and injury event (Table 4), initial concussion presentation (Table 5), and post-concussion outcomes (Table 6). Immediate removal from play was more prevalent in the MDQ + group ($p = 0.009$) with student-athletes that were immediately removed having 2.4 (95% CI 1.2, 4.9) greater odds of being MDQ + (Fig. 1). The MDQ + groups had lower prevalence of delayed onset of symptoms after injury ($p = 0.022$), with student-athletes with delayed symptom onset having 57% (OR: 0.43, 95% CI 0.2, 0.9) lower odds of being MDQ + (Fig. 1).

Post-concussion alcohol ($p = 0.020$), tobacco ($p = 0.048$), and marijuana ($p = 0.026$) use were more prevalent in the MDQ + group. Alcohol, tobacco, and marijuana use since injury increased the odds of MDQ by 3.5 (95% CI 1.1, 9.5), 5.4 (95% CI 1.5, 19.0), and 2.6 (95% CI 1.2, 5.4), respectively (Fig. 1). On average, the MDQ + group took longer to become asymptomatic (MDQ +: 23.5 days, 95% CI 15.8, 31.2 vs. MDQ−: 10.6 days, 95% CI 9.5, 11.6; $p < 0.001$), and had a greater BSI 18 anxiety subscore at 24–48 h ($p = 0.011$). Figure 2 contains a survival curve for MDQ status across days to asymptomatic and shows that most MDQ decisions are made prior to 100 days post-injury. Survival curve analyses across factors associated with MDQ are presented as electronic supplemental material. It should be noted that days to asymptomatic was disproportionately missing in the MDQ + group (24.5%) compared to the MDQ− group (9.1%), likely because the higher rates or prolonged recovery in the MDQ + group make it more difficult to track exact

Table 1 Demographic and scholastic factors for medical disqualified (MDQ+) and non-medical disqualified (MDQ−) concussed student-athletes

	p	MDQ+ *N*=53 (2.9%)		MDQ MDQ− *N*=1779 (97.1%)	
		% within group	*N*/Total within group	% within group	N/total within Group
Sex	0.957				
Female		41.5	22/53	41.9	745/1779
Male		58.5	31/53	58.1	1034/1779
Ethnicity	0.456				
Hispanic/Latino		3.8	2/53	8.0	143/1779
Not Hispanic/Latino		79.2	42/53	78.2	1391/1779
Unknown		17.0	9/53	13.8	245/1779
Race	0.389				
American Indian/Alaska Native		1.9	1/53	0.4	8/1779
Asian		0.0	0/53	1.9	33/1779
Native Hawaiian/Pacific Islander		0.0	0/53	0.8	15/1779
African American/Black		13.2	7/53	19.4	345/1779
White		77.4	41/53	66.0	1175/1779
Unknown		1.9	1/53	3.4	60/1779
Multiple races		5.7	3/53	8.0	143/1779
	p	MDQ+ *N*=53 (2.9%)		MDQ MDQ− *N*=1779 (97.1%)	
		Mean	SD	Mean	SD
Age (years)	0.455	19.4	1.2	19.3	1.3
Cumulative college GPA	0.457	3.13	0.40	3.06	0.46
Cumulative high school GPA (4 pt scale)	0.817	3.58	0.44	3.62	0.47
SAT (on 2400 scale)	0.542	1724.2	206.4	1762.2	254.4
ACT	0.466	26.2	4.7	25.8	4.5
	p	MDQ+ *N*=53 (2.9%)		MDQ MDQ− *N*=1779 (97.1%)	
		% within group	*N*/total within group	% within group	*N*/total within Group
Type of student	0.698				
Above Average		67.3	35/52	66.6	1103/1656
Average		30.8	16/52	32.5	539/1656
Below average		1.9	1/52	0.8	14/1656
Skipped a year/grade of school?-yes	1.000	1.9	1/53	1.9	34/1755
Repeated a year in school?-yes	0.535	7.8	4/51	5.7	100/1741
Received school mandated academic assistance?-yes	0.562	7.5	4/53	6.1	108/1779

days to asymptomatic. The MDQ+ group also tended to have a higher percentage of transport to hospital ($p=0.099$).

Per clinician report following concussion, the MDQ+ group had higher percentages of prolonged recovery due to: anxiety ($p=0.001$), depression ($p<0.001$), malingering ($p=0.014$), motivation ($p<0.001$), other injuries ($p=0.031$), and symptom exaggeration ($p=0.027$). Prolonged recovery due to secondary gains (advantage that occurs secondary to stated or real illness [10]) trended towards higher percentages in the MDQ+ group ($p=0.084$). The MDQ+ group experienced higher percentages of post-concussion psychiatric issues ($p=0.005$). For odds ratio calculations, we collapsed anxiety, depression, and psychiatrics issues into a single category: "prolonged recovery due to mental health alterations". Similarly, we grouped malingering, motivation, and symptom

Table 2 Sport factors for medical disqualified (MDQ+) and non-medical disqualified (MDQ-) concussed student-athletes

	p	MDQ+ *N* = 53 (2.9%)		MDQ- *N* = 1779 (97.1%)	
		% within group	*N*/total within group	% within group	*N*/total within group
Primary sport contact level	0.148				
Contact		81.1	43/53	69.8	1242/1779
Limited contact		9.4	5/53	19.8	352/1779
Non-contact		9.4	5/53	10.4	185/1779
Primary Sport	Not analyzed due to sample size				
Baseball		0.0	0/53	2.8	50/1779
Basketball		0.0	0/53	8.1	144/1779
Cheerleading		3.8	2/53	2.6	47/1779
Cross country/track		0.0	0/53	3.2	57/1779
Diving		0.0	0/53	1.6	29/1779
Fencing		0.0	0/53	0.3	6/1779
Field event		1.9	1/53	1.3	23/1779
Field hockey		5.7	3/53	1.4	25/1779
Football		47.2	25/53	28.6	509/1779
Golf		1.9	1/53	0.3	6/1779
Gymnastics		3.8	2/53	3.1	56/1779
Ice hockey		1.9	1/53	2.5	44/1779
Lacrosse		5.7	3/53	6.8	121/1779
Rifle		0.0	0/53	0.2	4/1779
Rowing/crew		1.9	1/53	2.0	35/1779
Soccer		15.1	8/53	14.7	262/1779
Softball		1.9	1/53	3.0	54/1779
Swimming		1.9	1/53	3.5	62/1779
Tennis		0.0	0/53	1.7	30/1779
Volleyball		1.9	1/53	5.7	102/1779
Volleyball (beach)		0.0	0/53	0.2	4/1779
Water polo		3.8	2/53	2.8	49/1779
Wrestling		1.9	1/53	3.3	59/1779
Sailing		0.0	0/53	0.1	1/1779
Division	0.200				
I		71.7	38/53	79.3	1410/1779
II		11.3	6/53	11.1	198/1779
III		17.0	9/53	9.6	171/1779
Expect to be a starter?-Yes	0.093[b]	54.5	18/33	68.4	854/1249
Where do you anticipate you'll be on the depth chart in the upcoming season?	0.077[a]				
1st String		39.0	16/41	55.1	661/1199
2nd String		36.6	15/41	30.6	367/1199
3rd String		24.4	10/41	14.3	171/1199
		Mean	SD	Mean	SD
Years of being a starter?	0.082[a]	8.1	3.9	9.1	4.1

[a]Trending difference between MDQ+ and MDQ− ($p = 0.05$–0.10)

exaggeration into a single "motivation" category. Prolonged recovery due to mental health alterations, motivation, and other injuries increased the odds of MDQ by 5.3 (95% CI 2.0, 14.1), 7.5 (95% CI 3.3, 17.0), and 3.7 (95% CI 1.3, 10.9), respectively (Fig. 1).

Table 3 Medical history factors for medical disqualified (MDQ+) and non-medical disqualified (MDQ−) concussed student-athletes

	p	MDQ+ *N* = 53 (2.9%)		MDQ− *N* = 1779 (97.1%)	
		% within group	*N*/total within group	% within group	*N*/total within group
Previous concussions	< 0.001[a]				
0		32.1	17/53	48.8	855/1752
1		24.5	13/53	36.5	640/1752
2		20.8	11/53	10.0	175/1752
3+		22.6	12/53	4.7	82/1752
	p	MDQ+ *N* = 53 (2.9%)		MDQ− *N* = 1779 (97.1%)	
		Mean	SD	Mean	SD
Average days to asymptomatic for previous concussion	0.142	14.9	16.3	10.6	19.0
	p	MDQ+ *N* = 53 (2.9%)		MDQ− *N* = 1779 (97.1%)	
		% within group	*N*/total within group	% within group	*N*/total within group
One or more headaches in past 3 months? Yes	0.048[a]	48.0	24/50	33.9	583/1721
Psychiatric disorder? Yes	1.000	3.9	2/52	3.9	67/1740
Learning disorder? Yes	0.716	3.8	2/52	3.7	64/1747
Attention deficit-hyperactivity disorder? Yes	0.795	5.8	3/52	8.2	143/1741
Headache disorder (non-migraine)? Yes	1.000	1.9	1/52	2.1	36/1746
Migraine headaches? Yes	0.118	13.5	7/52	7.6	133/1744
Alcohol use in the past month? Yes	0.231	47.9	23/48	39.3	565/1437
Tobacco use in the past month? Yes	0.391	11.8	6/51	8.4	146/1744
Marijuana use in past month? Yes	0.615	4.3	2/47	3.0	42/1410
	p	MDQ+ *N* = 53 (2.9%)		MDQ− *N* = 1779 (97.1%)	
		Mean	SD	Mean	SD
Brief sensation seeking scale total score	0.050[b]	3.3	0.6	3.1	0.7
Baseline symptom severity	0.573	4.9	8.2	5.0	9.1
Baseline BSI 18 Somatization subscore	0.590	0.64	1.39	0.90	1.95
Baseline BSI 18 Anxiety subscore	0.593	0.81	2.03	0.85	1.98
Baseline BSI 18 Depression subscore	0.618	1.08	2.18	0.87	2.12

[a]Significant difference between MDQ+ and MDQ− ($p < 0.05$)

[b]Trending difference between MDQ+ and MDQ− ($p = 0.05$–0.10)

4 Discussion

This study is the first exploratory analysis of MDQ following concussion in collegiate student-athletes and the first to detail an extensive prospectively collected list of premorbid and postmorbid factors. MDQ following concussion was relatively rare, present in just 2.9% of the concussed sample (53 student-athletes across 29 sites over four years). We identified three interesting patterns related to MDQ following concussion in our results (1) premorbid factors, aside from concussion and headache history, did not strongly associate with MDQ status, (2) initial concussion presentation was more severe in the MDQ+ group, and (3) postmorbid outcomes were most associated with MDQ status.

Table 4 Injury event factors for medical disqualified (MDQ+) and non-medical disqualified (MDQ−) concussed student-athletes

	p	MDQ+ *N* = 53 (2.9%)		MDQ− *N* = 1779 (97.1%)	
		% within group	*N*/total within group	% within group	*N*/total within group
Injury occurred during:	0.253				
Practice/training		66.0	35/53	56.2	998/1776
Competition		26.4	14/53	29.4	522/1776
Outside of sport		7.5	4/53	14.4	256/1776
Mechanism of injury (check all)					
Tackled by opponent	0.591	5.7	3/53	6.1	109/1779
Tackling an opponent	0.441	5.7	3/53	3.6	64/1779
Blocking	0.528	7.5	4/53	5.3	95/1779
Collision with opponent	0.730	22.6	12/53	20.5	365/1779
Collision with teammate	0.853	15.1	8/53	17.2	306/1779
Header with physical contact	0.400	0.0	0/53	2.9	51/1779
Header without physical contact	1.000	0.0	0/53	1.2	21/1779
Being checked	0.318	1.9	1/53	0.7	12/1779
Checking	1.000	0.0	0/53	0.1	2/1779
Contact with ball/puck/stick	0.558	17.0	9/53	14.6	259/1779
Other	0.296	24.5	13/53	32.4	577/1779
Athlete collided with (check all)					
Other player's head/helmet	0.873	26.4	14/53	25.4	451/1779
Other player's body	0.750	22.6	12/53	25.7	458/1779
Ground/floor/ice	0.134	24.5	13/53	16.5	293/1779
Field equipment	0.453	5.7	3/53	3.8	67/1779
Sideline equipment	1.000	0.0	0/53	0.2	3/1779
Other	0.228	22.6	12/53	31.5	560/1779
Impact location (check all)					
Head-front	0.885	35.8	19/53	34.9	621/1779
Head-side	0.303	26.4	14/53	34.1	607/1779
Head-back	0.293	24.5	13/53	19.0	338/1779
Head-top	0.178	1.9	1/53	7.6	136/1779
Trunk/torso	0.376	1.9	1/53	0.8	15/1779
Unknown	0.193	13.2	7/53	7.9	141/1779
Other	0.770	3.8	2/53	6.4	113/1779

[a]Significant difference between MDQ+ and MDQ− ($p < 0.05$)

[b]Trending difference between MDQ+ and MDQ− ($p = 0.05$–0.10)

4.1 Premorbid Factors

With the exception of concussion and headache history, premorbid factors largely did not associate with MDQ status. This contrasts with Tribassi who found that high school girls had twice the rate of MDQ following concussion than boys in sex-comparable sports [2]. Similarly, MDQ cases occurred evenly between event types, sports, and injury mechanism, but has previously been attributed to injuries sustained in competitions as a result of player-to-player contact [2].

Those with a concussion history or reporting one or more headaches in the past 3 months were more likely to be MDQ+. Concussion history has long been linked to MDQ, with longstanding myths that MDQ is imminent following three concussions. Our results show that MDQ odds are greater after three concussions, but that this was not an automatic MDQ criterion [11]. A considerable number of athletes were MDQ+ with a history of 2 or even 1 prior concussion, with others having a history of 3+ and remaining MDQ−. The National Athletic Trainers' Association position statement on disqualifying conditions suggests that athletes with a history of 3 or more concussions or

Table 5 Concussion presentation factors for medical disqualified (MDQ+) and non-medical disqualified (MDQ−) concussed student-athletes

	p	MDQ+ *N* = 53 (2.9%)		MDQ− *N* = 1779 (97.1%)	
		% within group	*N*/total within group	% within group	*N*/total within group
Immediately removed from play? – Yes	0.009[a]	74.4	32/43	54.5	841/1544
Return to play same day? – Yes	0.861	21.2	11/52	20.1	353/1753
Delay onset of symptoms after injury? -Yes	0.022[a]	17.6	9/51	33.3	578/1736
Alternation in mental status? – Yes	0.671	52.9	27/51	49.4	863/1747
Loss of consciousness? -Yes	1.000	3.8	2/52	5.6	99/1763
Post-traumatic Amnesia?—Yes	0.474	11.8	6/51	9.4	165/1761
Retro grade Amnesia? – Yes	0.182	9.8	5/51	5.0	88/1761
Demonstrate Motor Impairment? – Yes	0.621	15.4	4/26	19.3	186/966
Transport to hospital? – Yes	0.099[b]	13.5	7/52	7.1	126/1763
Alcohol used since injury? –Yes	0.020[a]	40.0	12/30	20.7	232/1122
Tobacco used since injury? –Yes	0.048[a]	11.4	4/35	3.8	55/1441
Marijuana used since injury? –Yes	0.026[a]	8.1	3/37	1.6	21/1308
Prescription drugs since injury? – Yes	0.228	11.6	5/43	7.0	119/1694
Over the counter medication since injury? – Yes	0.644	51.2	22/43	55.0	928/1686

[a]Significant difference between MDQ+ and MDQ− ($p < 0.05$)

[b]Trending difference between MDQ+ and MDQ− ($p = 0.05$–0.10)

delayed recovery may need to be considered for temporary or permanent disqualification from contact sports [12]. It is important to note that 32.1% of the MDQ+ group were disqualified with just one prior concussion (two total concussions including the studied case). This finding underscores the value of clinical judgement in guiding MDQ decisions and reinforces the fact that there is no set number of concussions that should force MDQ. Our results are similar to those of Welton et al., who showed that high school athletes with recurrent sport-related injuries were more likely to be medical disqualified or voluntarily withdraw from sport [13].

It is not immediately clear why one or more headaches in the past 3 months at baseline was associated with being MDQ+ (MDQ+ = 48.0% vs MDQ− = 33.9%) while diagnosed headache disorder and migraine history were not. It is possible that this medical history indicator is precipitous to a longer and more complicated concussion recovery, thus increasing likelihood of MDQ.

We also observed trends across variables related to a student-athlete's athletic achievement. Student-athletes with more established roles on their teams as starters had trends toward lower prevalence of MDQ. In contrast, NCAA division (I, II, III) was not associated with MDQ suggesting that practices do not vary largely across levels of play. Combined, these results suggest the level of play (i.e. NCAA division) may not hold as much influence as their local impact on team's success as a starter. In a survey-based study, clinicians that more strongly prioritized athletes' on-field achievements indicated that they would discuss MDQ after a larger number of concussions compared to those who less strongly value athletic achievement [14]. Interestingly, this tolerance to concussion history was not influenced by whether the clinician worked in a "power conference" or "non-power conference" university. The most recent AMSSM position statement does identify "perception of the benefit of sport participation (e.g., personal identify, financial motivation)" as a consideration for process of shared decision MDQ making [3]. Regardless of a student-athletes desire to continue sport participation, clinicians must objectively evaluate the risks of continued sport participation and must candidly discuss potential future risks with student-athletes throughout the MDQ decision-making process [1].

It is important to note that the outcomes of this study reflect the clinical presentation of NCAA student-athletes and the clinical practices of NCAA sports medicine professionals, but not necessarily best-practices for MDQ following concussion. We must better understand the immediate and long-term neurological consequences of each of these factors to discern whether factors should be criteria for MDQ. However, the results of this study guide future efforts to inform MDQ practices by displaying risk factors and current practices.

4.2 Initial Concussion Presentation

Athletes that were MDQ+ had more frequent immediate removal from play, less frequent delayed symptom onset,

Table 6 Post-concussion outcomes for medical disqualified (MDQ+) and non-medical disqualified (MDQ−) concussed student-athletes

	p	MDQ+ *N* = 53 (2.9%)		MDQ− *N* = 1779 (97.1%)	
		Mean	SD	Mean	SD
Days to asymptomatic	< 0.001[a]	23.5	24.0	10.6	21.8
Total Symptom Score @ 0–6 h	0.328	32.1	24.3	27.47	21.1
Total Symptom Score @ 24–48 h	0.085[b]	29.5	22.5	24.3	21.2
Total Symptom Score @ Asymptomatic	0.291	2.3	4.4	2.1	5.9
BSI 18 Somatization subscore @ 24–48 h	0.207	3.4	3.7	2.6	3.0
BSI 18 Somatization subscore @Asymptomatic	0.967	0.7	1.4	0.7	1.5
BSI 18 Anxiety subscore @ 24–48 h	0.011[a]	2.6	3.6	1.4	2.5
BSI 18 Anxiety subscore @ Asymptomatic	0.456	0.5	1.4	0.4	1.2
BSI 18 Depression subscore@ 24–48 h	0.147	1.9	2.9	1.5	2.6
BSI 18 Depression subscore @ Asymptomatic	0.442	0.6	1.9	0.5	1.5
	p	MDQ+ *N* = 53 (2.9%)		MDQ− *N* = 1779 (97.1%)	
		% within group	*N*/total within group	% within group	*N*/total within group
Experience psychiatric issues?—Yes	0.005[a]	11.6	5/53	2.4	41/1779
Prolonged recovery due to: (check all)					
Academic requirements—Yes	0.159	5.7	3/53	2.5	45/1779
Anxiety—Yes	0.001[a]	15.1	8/53	3.3	59/1779
Depression—Yes	< 0.001[a]	13.2	7/53	1.2	23/1779
Malingering—Yes	0.014[a]	5.7	3/53	0.8	15/1779
Motivation—Yes	< 0.001[a]	15.1	8/53	1.5	27/1779
Other injuries—Yes	0.031[a]	7.5	4/53	2.1	38/1779
Physical activity—Yes	1.000	1.9	1/53	2.1	37/1779
Secondary gains—Yes	0.084[b]	1.9	1/53	0.1	2/1779
Somatization—Yes	1.000	0.0	0/53	0.2	3/1779
Stress—Yes	0.307	7.5	4/53	4.6	81/1779
Symptom exaggeration—Yes	0.027[a]	5.7	3/53	1.1	20/1779

[a]Significant difference between MDQ+ and MDQ− ($p < 0.05$)

[b]Trending difference between MDQ+ and MDQ− ($p = 0.05$–0.10)

less frequent delays in symptom onset, higher BSI 18 anxiety subscores at 24–48 h following injury, and tended to have a higher percentage of transport to hospital. However, other acute markers, such as loss of consciousness, altered mental status, and amnesia had equal prevalence between groups. Concussions resulting in MDQ may have a more severe initial presentation with immediate signs and symptoms, warranting immediate removal. However, immediate removal should not be considered predictive of MDQ as immediate removal from play has previously been linked to shorter recovery duration [15, 16]. Unfortunately, we were not able to assess whether the MDQ+ group had higher prevalence of structural abnormality on neuroimaging due to low occurrence of imaging in both groups. Furthermore, we were not able to determine whether the MDQ+ group had reduced threshold of sustaining recurrent concussions, which has previously been suggested as a criterion for MDQ following concussion.

4.3 Postmorbid Outcomes

The MDQ+ group took an average of 12.9 more days to become asymptomatic. Our survival curve shows an interesting pattern where MDQ occurred progressively up until ~50 days asymptomatic, but a notable increase in MDQ percentage occurred between days 50 and 100 (Fig. 2). It

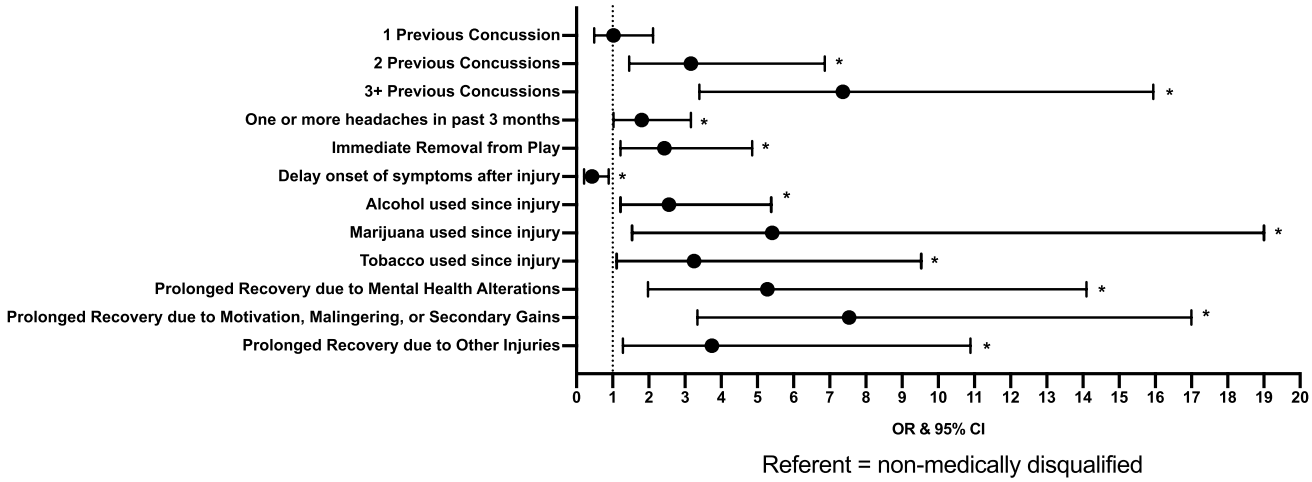

Fig. 1 Forest plot with odds ratios (and corresponding 95% confidence intervals) for medical disqualification (MDQ) following concussion. Odds ratios > 1 express increased odds of MDQ in the presence of the y axis factor. * The 95% confidence interval does not include 1

is possible that some of the major risk factors associated with MDQ encourage the MDQ decision early, regardless of the recovery duration, but that a combination of prolonged recovery with associated risk factors expedites this decision-making. In addition to longer recoveries, MDQ + student-athletes experienced more complicated recoveries with three primary post-concussion presentations: mental health alterations, substance use, and motivation issues. We found that 11.6% of MDQ + group—compared with just 2.4% of the MDQ− group—experienced some form of post-concussion mental health alteration that was attributed to their prolonged recovery. In contrast, premorbid factors of psychiatric or mood disorders were not associated with being MDQ +, suggesting that either the concussion, the MDQ process, or the combination of both triggered the mental health alterations. Post-concussion mental health alterations are common [17], with feelings of depression and irritability being present in 20.4% and 20.2% of concussion cases at 6 weeks following the injury, respectively [18]. For some student-athletes, MDQ may carry an additional mental health burden, because the disqualification removes them from their team social system and potentially threatens their longstanding athletic identity [19]. However, we cannot determine whether the post-concussion mental health alterations were precipitous to or a result of the MDQ decision. Furthermore, athlete stigma towards premorbid self-reported mental health alterations may influence their baseline report accuracy [20], which could differ considerably from the postmorbid clinical reported outcomes observed in this study. Clinicians medically disqualifying a student-athlete should consider and manage the potential mental health consequences that may precede or follow a MDQ decision, especially in the presence of mental health alterations present following concussion.

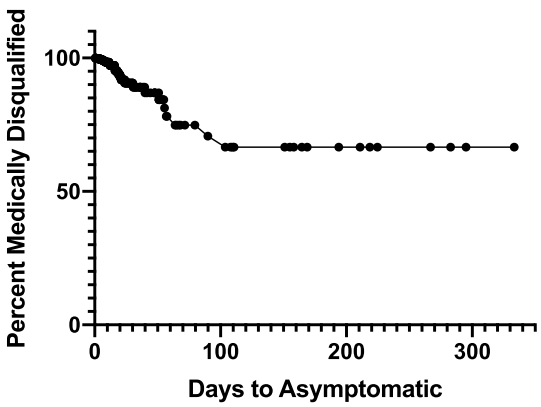

Fig. 2 Survival curve for overall medical disqualification across days to asymptomatic

Post-concussion substance use, specifically alcohol, tobacco, and marijuana were more common in the MDQ + group and was associated with greater odds of MDQ. This highlights the need to closely monitor substance use following concussion. It is possible that post-concussion substance use could be an attempt to self-medicate by the student-athlete, which could be tied to the post-concussion symptom burden, emotional burden of impending MDQ, or motivational issues. It is also possible that substance use could be related to the mental health consequences of concussion or may influence a clinician's willingness to medically disqualify an athlete. Premorbid substance use was not related to MDQ status, but these results should be interpreted with caution as student-athletes are unlikely to honestly report illegal activities related to underage drinking or illegal substance use.

Most concussed individuals experience complete recovery, however, a number experience persistent symptoms that may appear inconsistent with the severity of the injury to a treating clinician. Our results showed that clinician-reported motivation, malingering, and symptom exaggeration following concussion were strongly associated with increased odds of MDQ. Though interesting, these findings are difficult to interpret and only represent the clinician's impression, which could be misattributed as often occurs in traumatic brain injury and mental health management [21, 22]. Motivation, symptom exaggeration, and malingering are complex issues and may be associated with other complex differential diagnoses. Treatment options, such as cognitive behavioral therapy [23], are available to reduce prolonged concussion recovery due to poor motivation, symptom exaggeration and malingering, and should be explored prior to a final MDQ decision [21, 22]. In addition, it is important to recognize that athletes recovering from a concussion who appear to lack motivation, malinger or exaggerate symptoms could be seeking a respectable reason to leave or quit a team and may see the injury as a way out. This is also supported by our findings regarding secondary gains. Student-athletes may occasionally pursue MDQ following concussion as a means to secure a scholarship without having to continue sport participation. All of these post-injury outcomes are rated by managing clinicians associated with the CARE Consortium, thus, it is important to note that these risk factors reflect the clinician's impression which may not accurately represent the student-athletes true state.

This study had limitations. We captured formal medical disqualification, but medical disqualification is not this simplistic. A student-athlete with no remaining athletic eligibility that sustains a concussion at the last game of the season may qualify for MDQ, but may not be recorded since sport obligations have resolved. Conversely, student-athletes and clinicians may have other reasons to pursue MDQ that precipitate a concussion, for which the concussion acts as a catalyst to begin the formal process. We grouped MDQ + athletes across all clinician responses that indicated that the student-athlete was disqualified for any amount of time. However, MDQ decisions are more fluid than this dichotomous outcome. The response options did not allow us to fully determine the finality of the MDQ decision (i.e. for the season vs. career disqualification). Permanent MDQ decisions are typically much more complex and challenging than MDQ for the season. Further MDQ for the season often precipitates a more permanent MDQ decision. Due to the rarity of MDQ, our sample size of 53 MDQ + cases is relatively small, prohibiting us from building more comprehensive models with confounding factors, but represents the largest sample to date with so many descriptive variables. One such factor is risk for prolonged recovery, which warrants further study. The CARE Consortium is a large multi-site study aimed at investigating the natural history of concussion which limited our ability to link premorbid and postmorbid factors to MDQ. More targeted studies are necessary to fully capture this complex decision-making process. Furthermore, MDQ decisions are meant to protect student-athletes from long-term consequences, which are not captured within the CARE Consortium. Guidelines for MDQ are scarce and largely based on expert opinion. Clinician practices likely vary greatly across sites and we have no record of site-specific clinical practices. Future research should assess MDQ practices and opinions among managing clinicians.

5 Conclusion

Medical disqualification after concussion was considerably rare in this large-scale study of NCAA student-athletes. By comparing pre- and postmorbid factors, we identified several factors associated with medical disqualification following concussion. Overall, medical disqualification following concussion was more prevalent in athletes with more extensive concussion histories, more frequent premorbid headaches, more severe initial presentations, that then experienced a longer and more complicated recovery. However, our findings further highlight that each concussion case presents and is treated individually and that no strict parameters can currently be applied by all clinicians when making difficult MDQ decisions.

Acknowledgements CARE Consortium Investigators (institution): April Hoy MS, ATC, CSCS (School of Behavioral and Applied Sciences, Azusa Pacific University). Joseph Hazzard, Jr., PhD, ATC (Department of Exercise Science, Bloomsburg University). Louise Kelly, PhD (Department of Exercise Science, California Lutheran University). Christina Master, MD (Division of Orthopedics, Children's Hospital of Philadelphia). Justus Ortega, PhD (Department of Kinesiology & Recreation Administration, Humboldt State University). Nicholas Port, PhD (School of Optometry, Indiana University). Darren Campbell MD, Col USAF (retired) (Intermountain Sports Medicine). Steven J. Svoboda, MD, Colonel, US Army(retired) (MedStar Orthopaedic Institute). Margot Putukian, MD (Athletic Medicine, Princeton University). Sara P.D. Chrisman MD, MPH (Center for Child Health, Behavior and Development, Seattle Children's Research Institute). Dianne Langford, PhD (Lewis Katz School of Medicine, Temple University). Gerald McGinty, PT, DPT (United States Air Force Academy). Jonathan C. Jackson MD (United States Air Force Academy). Kenneth L. Cameron, PhD, MPH, ATC, FNATA (Keller Army Hospital, United States Military Academy). Megan N. Houston, PhD, ATC (Keller Army Hospital, United States Military Academy). Adam James Susmarski, DO (Department Head Brigade Orthopaedics and Sports Medicine, United States Naval Academy). Joshua T. Goldman, MD, MBA (Departments of Family Medicine & Orthopaedic Surgery, University of California, Los Angeles). Christopher Giza, MD (Department of Pediatrics, University of California, Los Angeles). Holly Benjamin, MD (Department of Rehabilitation Medicine and Pediatrics, University of Chicago). Thomas Buckley, PhD (Department of Kinesiology & Applied Physiology, University of Delaware). Thomas Kaminski, PhD, ATC (Department of Kinesiology & Applied Physiology, University of. Delaware).

Luis Feigenbaum, DPT, ATC (Department of Physical Therapy, Miller School of Medicine, University of Miami). James T. Eckner, M.D., M.S. (Department of PM&R, University of Michigan). Jason P. Mihalik, PhD, CAT(C), ATC, FACSM (Department of Exercise and Sport Science, The University of North Carolina at Chapel Hill). Scott Anderson ATC (University of Oklahoma). Anthony Kontos, PhD (Department of Orthopaedic Surgery, University of Pittsburgh). M. Alison Brooks, MD, MPH (Department of Orthopedics, University of Wisconsin, Madison). Stefan M. Duma, PhD (Institute for Critical Technology and Applied Science, Virginia Tech). Steve Rowson, PhD (Department of Biomedical Engineering and Mechanics, Virginia Tech). Christopher Miles, MD (Department of Family and Community Medicine, Wake Forest University). Laura Lintner, DO (Wake Forest School of Medicine Family Medicine, Winston Salem State University).

Author Contributions JDS, MLWR, and RCL contributed to the conception and design of this study. All authors contributed to the acquisition, analysis, and interpretation of data. JDS drafted the manuscript, which was then revised critically for important intellectual content by all authors. All authors granted final approval of the version to be published and agree to be accountable for all aspects of the work.

Data Availability Statement Data are available within the Federal Interagency Traumatic Brain Injury Research (FITBIR) informatics system at https://fitbir.nih.gov/.

Compliance with Ethical Standards

Funding This publication was made possible, in part, with support from the Grand Alliance Concussion Assessment, Research, and Education (CARE) Consortium, funded, in part by the National Collegiate Athletic Association (NCAA) and the Department of Defense (DOD). The U.S. Army Medical Research Acquisition Activity, 820 Chandler Street, Fort Detrick MD 21702-5014 is the awarding and administering acquisition office. This work was supported by the Office of the Assistant Secretary of Defense for Health Affairs through the Psychological Health and Traumatic Brain Injury Program under Award NO W81XWH-14-2-0151. Opinions, interpretations, conclusions and recommendations are those of the author and are not necessarily endorsed by the Department of Defense (DHP funds). This manuscript was also made possible, in part, with support from the Air Force Research Laboratory's Summer Faculty Fellowship Program.

Conflict of Interest Julianne Schmidt, Michelle Weber Rawlins, James Clugston, Steven Broglio, Michael McCrea, and Thomas McAllister received funding from the National Collegiate Athletic Association and the Department of Defense to complete this investigation and cover travel costs related to the study. The remaining authors declare that they have no conflicts of interest.

References

1. Davis-Hayes C, Baker DR, Bottiglieri TS, et al. Medical retirement from sport after concussions: a practical guide for a difficult discussion. Neurol Clin Pract. 2018;8(1):40–7.
2. Tirabassi J, Brou L, Khodaee M, et al. Epidemiology of high school sports-related injuries resulting in medical disqualification. Am J Sports Med. 2016;44(11):2925–32.
3. Harmon KG, Clugston JR, Dec K, et al. American Medical Society for Sports Medicine position statement on concussion in sport. Br J Sports Med. 2019;53(4):213–25.
4. Cantu RC. Recurrent athletic head injury: risks and when to retire. 2003, W B SAUNDERS CO: United States. p. 593.
5. Sedney CL, Orphanos J, Bailes JE. When to consider retiring an athlete after sports-related conclusion. Clin Sports Med. 2011;30:189–200.
6. Huml MR. A Factor Structure Examination of Athletic Identity Related to NCAA Divisional Differences. J Coll Stud Dev. 2018;59(3):376–81.
7. Broglio SP, McCrea M, McAllister T, et al. A national study on the effects of concussion in collegiate athletes and US military service academy members: the NCAA-DoD concussion assessment, research and education (CARE) consortium structure and methods. Sports Med. 2017;47(7):1437–51.
8. Hoyle RH, Stephenson MT, Palmgreen P, et al. Reliability and validity of a brief measure of sensation seeking. Pers Individ Dif. 2002;32:401–14.
9. Broglio SP, AP Kontos, H Levin, et al. National Institute of Neurological Disorders and Stroke and Department of Defense Sport-Related Concussion Common Data Elements Version 1.0 Recommendations. *J Neurotrauma.* 2018;35(23):2776–2783.
10. Davidhizar R. The pursuit of illness for secondary gain. Health Care Superv. 1994;13(1):10–5.
11. Harmon KG, Drezner J, Gammons M, et al. American medical society for sports medicine position statement: concussion in sport. Clin J Sport Med. 2013;23(1):1–18.
12. Conley KM, Bolin DJ, Carek PJ, et al. National Athletic Trainers' Association position statement: Preparticipation physical examinations and disqualifying conditions. J Athl Train. 2014;49(1):102–20.
13. Welton KL, Kraeutler, MJ, Pierpoint, LA, et al. Injury recurrence among high school athletes in the United States: a decade of patterns and trends, 2005–2006 through 2015–2016. Orthop J Sports Med 6(1):2325967117745788.
14. Kroshus E, Baugh CM, Meehan WP, et al. Personal subjectivity in clinician discussion about retirement from sport post-concussion. Soc Sci Med. 2018;218:37–44.
15. Asken BM, Bauer RM, Guskiewicz KM, et al. Immediate removal from activity after sport-related concussion is associated with shorter clinical recovery and less severe symptoms in collegiate student-athletes. Am J Sports Med. 2018;46(6):1465–74.
16. Elbin RJ, Sufrinko A, Schatz P, et al. Removal from play after concussion and recovery time. Pediatrics. 2016;138(3):1.
17. Rice SM, Parker AG, Rosenbaum S, et al. Sport-related concussion and mental health outcomes in elite athletes: a systematic review. Sports Med. 2018;48(2):447.
18. Chaput G, Giguère J-F, Chauny J-M, et al. Relationship among subjective sleep complaints, headaches, and mood alterations following a mild traumatic brain injury. Sleep Med. 2009;10(7):713–6.
19. Solomon GS, Kuhn AW, Zuckerman SL. Depression as a modifying factor in sport-related concussion: a critical review of the literature. Phys Sportsmed. 2016;44(1):14–9.
20. Kaier E, Cromer LD, Johnson MD, et al. Perceptions of mental illness stigma: comparisons of athletes to nonathlete peers. J Coll Stud Dev. 2015;56(7):735–9.
21. Silver JM. Effort, exaggeration and malingering after concussion. J Neurol Neurosurg Psychiatry. 2012;83(8):836–42.
22. Merckelbach H, Dandachi-FitzGerald B, van Helvoort D, et al. When Patients Overreport Symptoms: More Than Just Malingering. Curr Dir Psychol Sci. 2019;28(3):321–6.
23. Silverberg ND, Hallam BJ, Rose A, et al. Cognitive-behavioral prevention of postconcussion syndrome in at-risk patients: a pilot randomized controlled trial. J Head Trauma Rehabil. 2013;28(4):313–22.

Affiliations

Julianne D. Schmidt[1,2] · Michelle L. Weber Rawlins[3] · Robert C. Lynall[1,2] · Christopher D'Lauro[4] · James R. Clugston[5] · Tom W. McAllister[6] · Michael McCrea[7] · Steven P. Broglio[8,9] · CARE Consortium Investigators

Michelle L. Weber Rawlins
michelleweber@atsu.edu

Robert C. Lynall
rlynall@uga.edu

Christopher D'Lauro
chris.dlauro@usafa.edu

James R. Clugston
jayclug@ufl.edu

Tom W. McAllister
twmcalli@iupui.edu

Michael McCrea
mmccrea@mcw.edu

Steven P. Broglio
broglio@umich.edu

[1] UGA Concussion Research Laboratory, University of Georgia, 330 River Rd, Athens, GA, USA

[2] Department of Kinesiology, University of Georgia, 330 River Rd, Athens, GA, USA

[3] Department of Interdisciplinary Health Sciences, AT Still University, Mesa, AZ, USA

[4] Department of Behavioral Science and Leadership, United States Air Force Academy, USAF, Air Force Academy, CO, USA

[5] Department of Community Health and Family Medicine, University of Florida, Gainesville, FL, USA

[6] Department of Psychiatry, Indiana University School of Medicine, Indianapolis, IN, USA

[7] Department of Neurosurgery, Medical College of Wisconsin, Milwaukee, WI, USA

[8] Michigan Concussion Center, University of Michigan, Ann Arbor, MI, USA

[9] School of Kinesiology, University of Michigan, Ann Arbor, MI, USA